



State of the Services

FY 2024 Report

October 2023 - September 2024

Engagement Events Highlights

NOAA Alaska NextGen Satellite Workshop

Anchorage, AK | September 10-12

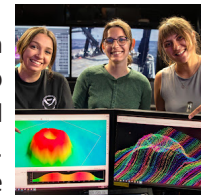
- NOAA's Alaska NextGen Satellite Workshop brought together scientists, state staff, and private sector partners to make decisions related to environmental monitoring, climate change adaptation, disaster preparedness, resource management, and policy development.
- This workshop highlighted Alaska's unique climate hazards and challenges, helped identify unfulfilled needs related to weather, water, and climate risks in Alaska's communities, and showcased the value of NOAA satellite products in the region.



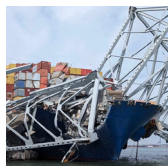
NOAA Ship *Okeanos Explorer* Mapping Cruise

Hawai'i | May 14 - June 9

- Members of NCEI participated in an ocean mapping cruise across Hawai'i to map previously unexplored regions and collect comprehensive data for the U.S. Bureau of Energy Management and the U.S. Geological Survey.
- During the mission they deepened their understanding of data acquisition, quality control, and interpretation processes. Their experience with sonar systems and data management tools improved efficiency of data workflows and expedited public access to high-quality marine geophysical data, supporting further oceanographic research.

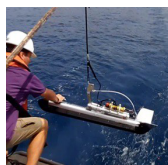


Product and Services Highlights



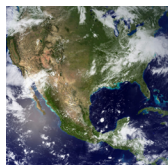
Bathymetric Attributed Grid (BAG) Data Viewer Mosaic Updated

NCEI's Bathymetric Data Viewer was updated with the latest BAG data this July. This quarterly update provided the latest visualization of derived bathymetric surfaces on the map interface and enhances user discoverability of National Ocean Service hydrographic survey data. This update included recent hydrography in the vicinity of the Key Bridge wreckage site at the Port of Baltimore. Users can observe seabed disturbance related to the response and reopening of the channel, with all collected data being hosted at NCEI.



Unlocking Underwater History

Sidescan sonar is a marine exploration and archaeology tool that provides detailed images of the seafloor to identify underwater features. The Aviators Down sidescan sonar project aimed to locate and identify Tuskegee and Free French World War II aircraft that crashed into Lake Huron, collecting data aboard the R/VI Storm. This project provided insights into World War II aviation history. The Lost Whaling Fleet project gathered data in search of the Lost Whaling Fleets in the Western Arctic, finding the remains of 2 of the 33 ships that were lost in the 1871 disaster. Sidescan sonar data from both projects have been archived and made publicly accessible.



Industry Proving Grounds (IPG)

The Inflation Reduction Act's IPG moved from planning to implementing in Spring FY 2024. IPG is a new effort to develop and share actionable climate information and improve information delivery to industry partners. NOAA, through NCEI, will work directly with the architecture, insurance, reinsurance, and retail industries to allow these sectors to better assess climate risk and empower rapid decision making and long-term planning. Through ongoing engagement with key members of these sectors and developing new, informational products like maps, reports, and datasets, NOAA's information will strengthen industry decision making related to climate resilience and help protect communities most vulnerable to climate change and its impacts.



Wind Climatology

The Western Regional Climate Center has developed a gridded wind climatology to support various sectors, including wildfire, agriculture, and energy. Utilizing the Unrestricted Mesoscale Analysis, the dataset includes wind speed, wind direction, and peak gust back to 1985. Other Regional Climate Centers (RCCs) have begun developing user interfaces for ease of access. One interface, developed by the High Plains RCC, includes options for selecting wind roses, gust hours and wind streaks.

Stakeholder Highlights



NCEI worked with the Office of Ocean Exploration and Research on an expedition in the Papahānaumokuākea Marine National Monument to expand mapping coverage in unexplored areas of the monument, particularly in waters deeper than 200 meters. Geophysical data was collected using continuous multibeam sonar and expendable bathythermographs. Conductivity, temperature, and depth measurements were used for environmental data, and water samples were collected for environmental DNA analysis.



Over the past four decades, the Arctic has experienced profound and unprecedented changes. While these shifts have allowed for new areas of commerce they have also caused increased intensity and frequency of coastal storms, among other repercussions. NCEI partnered with the National Snow and Ice Data Center to explore this vital area through data, resulting in the release of the *Monitoring the Melt: Measuring Arctic Sea Ice Loss* story-map.



NCEI collaborated with the National Integrated Heat Health Information System and over 60 other organizations to conduct extreme heat health tabletop exercises in Oklahoma City and Philadelphia. These exercises helped align more than 30 regional and local organizations in each location, enabling cities to increase their resilience to heat in ways that are unique to each location. NCEI also made significant contributions to the resources published in 2024 that are supporting the Heat Tabletop Exercise Planning Challenge-- resources based on lessons learned from the Climate Equity Listening Sessions in FY21, and the four cities exercised in FY22-FY23.



NOAA released the Equitable Climate Services Delivery Action Plan in May. The Plan is the culmination of a multi-year effort, during which NCEI worked with a cross-NOAA team to gather and assess public comments, and compile the information into actionable activities. The implementation of this Plan will help ensure that NCEI climate data, tools, and information advance climate resilience and decision support by improving climate services delivery to diverse audiences.

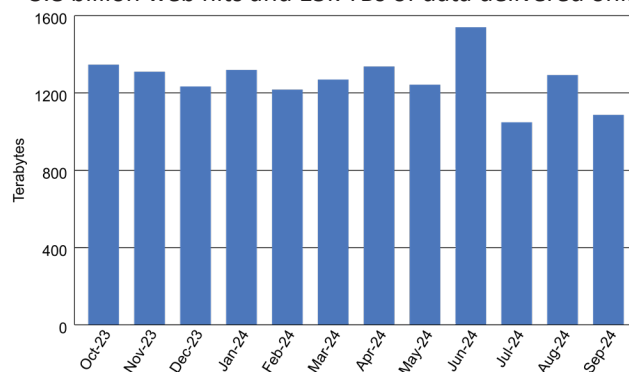
Upcoming Engagements

Climate Connections 2.0	Oct. 28 - 30
Seabed 2030 SW Pacific Mapping Meeting	Oct. 30 - Nov. 10
National Oceanographic Partnership Program Annual Meeting	Nov. 18 - 21

Customer Reach

NCEI Makes Data Accessible

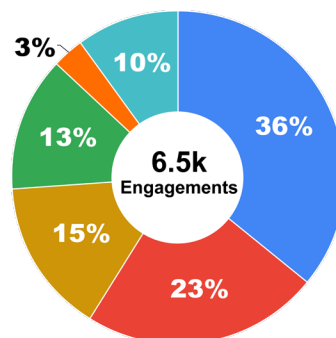
- 3.8 billion web hits and 15k TBs of data delivered online*



*Analytics include model & satellite data; data statistics may not be comprehensive across NCEI (e.g. CLASS)

NCEI Serves Every U.S. Sector in FY2024

- Top Sectors
- Professional, Scientific, and Technical Services
- Other Services (except Public Administration)
- Educational Services
- Public Administration
- Finance and Insurance
- Others (Combined)



Unique Customer Requests

Global Marine Microplastics Database Request

A cybersecurity and information technology company aims to develop an advanced image processing system to detect and quantify microplastics in oceanic waters. They will use the NCEI global Marine Microplastics datasets to train their machine-learning models and conduct comprehensive analyses. Their hope is that this initiative will contribute significantly to environmental monitoring and pollution control efforts.

Amazon Driver Training

An Amazon program manager who manages driver simulators used to train new Amazon drivers reached out to NCEI for winter weather data to determine the locations where their training would have the largest impact. NCEI data from the Global Historical Climatology Network-Daily and Local Climatology Data datasets, as well as the Midwestern Regional Climate Center's snowfall climatology tool, will all inform Amazon's Winter Weather Campaign for the 2024/2025 winter season.

Fans



129.6k



65.5k



16.7k

